

Java - both being interpretive languages, though Python is known for being easier to use. The vast availability of free libraries in Python, and its growing support in electronic systems like Raspberry Pi - coupled with its famed readability and 'Pythonic' simplicity, have lead to an increasing popularity for Python. On the other hand, Java's powerful inheritance structures and reusability of code provided by the Java Virtual Machine have made it a staple in certain applications of robotics.

The next bigwig is MATLAB, developed by Mathworks. Entire robotics systems have been developed with MATLAB alone, especially with the aid of its extremely well-supported and documented Robotics Toolkit. MATLAB's diversity and huge swath of inbuilt functions make it the language of choice in applications that need to process data, generate visuals, or implement advanced control systems.

Besides these, there are several Industrial Robotics Languages as well as Computer Aided Design/Modelling softwares that can be interfaced and used together for the design and simulation processes.

Behind the code - how robots learn

While the entire field of Artificial Intelligence is ultimately concerned with developing an autonomous intellect that can independently control a robot, work in the field is still scattered, albeit very steadily advancing in several interesting directions. Accordingly, as adapted from certain standard machine learning algorithms, there are three basic methodologies for a robot to learn how to behave.

- **Motor Babble:** Much like it sounds, motor babble is the use of a random input to each individual motor, coupled with data from various sensors used to set up a cost function that is then minimised over several iterations of parameters. After a variably lengthy learning period, the robot will be expected to have correlation information between the expected pattern of sensory feedback and the output signal that preceded it. For simpler systems where analytical solutions exist (for example, in inverse kinematical modules), this skip becomes redundant - most of the data is then determinate for various tweaked parameters.
- **Imitation:** Once a robot has a coordinated self-awareness, it can set about trying to imitate useful behaviours. This could be as simple as one robot monitoring the movement of another robot and mimicking it, or even as complicated as a quadcopter trying to ascertain the location of a thrown hoop so as to fly through it.